

Database System Engineering and Distributed Backend Development

Course Code: 25CS1302E

Week-1

2520030098

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Section - 3

Query 1: Create the database

```
1 • CREATE DATABASE IF NOT EXISTS bookflows_db;
```

Output:

```
✓ 1 20:24:43 CREATE DATABASE IF NOT EXISTS bookflows_db
```

Query 2: Select the database

```
2 • USE bookflows_db;
```

Output:

```
✓ 2 20:26:17 USE bookflows_db
```

Query 3: Create the books table

```
1 • CREATE DATABASE IF NOT EXISTS bookflows_db;
2 • USE bookflows_db;
3 • CREATE TABLE books (
4     book_id INT AUTO_INCREMENT PRIMARY KEY,
5     title VARCHAR(255) NOT NULL,
6     isbn VARCHAR(13) NOT NULL UNIQUE,
7     published_year INT,
8     CONSTRAINT chk_published_year CHECK (published_year < 2027)
9 );
10 • SELECT * FROM books;
```

Output:

```
✓ 3 20:28:11 CREATE TABLE books ( book_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(255) NOT NULL, i... 0 row(s) affected
```

	book_id	title	isbn	published_year
*	NULL	NULL	NULL	NULL

Query 4: Create the members table:

```

1 • CREATE DATABASE IF NOT EXISTS bookflows_db;
2 • USE bookflows_db;
3 • CREATE TABLE books (
4     book_id INT AUTO_INCREMENT PRIMARY KEY,
5     title VARCHAR(255) NOT NULL,
6     isbn VARCHAR(13) NOT NULL UNIQUE,
7     published_year INT,
8     CONSTRAINT chk_published_year CHECK (published_year < 2027)
9 );
10 • SELECT * FROM books;
11 • CREATE TABLE members (
12     member_id INT AUTO_INCREMENT PRIMARY KEY,
13     full_name VARCHAR(100) NOT NULL,
14     email VARCHAR(150) NOT NULL UNIQUE
15 );
16 • SELECT * FROM members;

```

Output:

	member_id	full_name	email
*	NULL	NULL	NULL

Query 5 (verification): Inspect the table structures

```
DESCRIBE books;
```

Output:

Field	Type	Null	Key	Default	Extra
book_id	int	NO	PRI	NULL	auto_increment
title	varchar(255)	NO		NULL	
isbn	varchar(13)	NO	UNI	NULL	
published_year	int	YES		NULL	

```
18 • DESCRIBE members;
```

Output:

	Field	Type	Null	Key	Default	Extra
▶	member_id	int	NO	PRI	NULL	auto_increment
	full_name	varchar(100)	NO		NULL	
	email	varchar(150)	NO	UNI	NULL	

Query 6: Insert 3 books

```
INSERT INTO books (title, isbn, published_year) VALUES
('The Alchemist', '9780061122415', 1988),
('Clean Code', '9780132350884', 2008),
('Atomic Habits', '9780735211292', 2018);
```

Output:

```
✓ 10 20:42:08 INSERT INTO books (title, isbn, published_year) VALUES ('The Alchemist', '9780061122415', 1988), ('Cle
```

Query 7: Insert 3 members

```
INSERT INTO members (full_name, email) VALUES
('Gutha Sushma Sree', 'sush@example.com'),
('Gutha Sowmya Sree', 'sow@example.com'),
('Gutha Mokshagna Naidu', 'moksha@example.com');
```

Output:

```
✓ 14 20:44:40 INSERT INTO members (full_name, email) VALUES ('Gutha Sushma Sree', 'sush@example.com'), ('Gutha Sov
```

```
SELECT * FROM books;
```

Output:

	book_id	title	isbn	published_year
▶	1	The Alchemist	9780061122415	1988
	2	Clean Code	9780132350884	2008
	3	Atomic Habits	9780735211292	2018
*	NULL	NULL	NULL	NULL

```
SELECT * FROM members;
```

Output:

	member_id	full_name	email
▶	1	Anil Kumar	anil.kumar@example.com
	2	Priya Sharma	priya.sharma@example.com
	3	Ravi Verma	ravi.verma@example.com

Constraint Tests (These MUST Fail):

Test 1: Duplicate ISBN (UNIQUE violation)

```
INSERT INTO books (title, isbn, published_year)
VALUES ('Fake Copy', '9780061122415', 2000);
```

Output:

```
✖ 26 09:10:01 INSERT INTO books (title, isbn, published_year) VALUES ('Fake Copy', '9780061122415', 2000)
```

Test 2: NULL title (NOT NULL violation)

```
INSERT INTO books (title, isbn, published_year)
VALUES (NULL, '9999999999999', 2010);
```

Output:

```
✖ 27 09:10:59 INSERT INTO books (title, isbn, published_year) VALUES (NULL, '9999999999999', 2010)
```

Test 3: Future publication year (CHECK violation)

```
INSERT INTO books (title, isbn, published_year)
VALUES ('Time Traveler', '8888888888888', 2030);
```

Output:

```
✖ 28 09:11:59 INSERT INTO books (title, isbn, published_year) VALUES ('Time Traveler', '8888888888888', 2030)
```

Test 4: Duplicate email (UNIQUE violation):

```
INSERT INTO members (full_name, email)
VALUES ('Anil Clone', 'anil.kumar@example.com');
```

Output:

```
✖ 29 09:13:04 INSERT INTO members (full_name, email) VALUES ('Anil Clone', 'anil.kumar@example.com')
```

-----X-----